TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	84.7 ± 1.68
Correlation Stability	92.5 ± 1.28
Distribution Consistency	90.3 ± 1.79
Entropy Preservation	89.0 ± 1.18
Mean Z Score Drift	0.1 ± 0.02
Categorical Fidelity	96.7 ± 1.79
Privacy Risk	100.0 ± 0.00
Data Density	99.9 ± 0.09
Coverage Score	0.0 ± 0.00
Feature Accuracy	92.3 ± 2.24
Novelty Score	0.0 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	84.7 ± 1.68
Correlation Stability (*2)	92.5 ± 1.28
Distribution Consistency (*3)	90.3 ± 1.79
Entropy Preservation (*4)	89.0 ± 1.18
Mean Z Score Drift (*5)	0.1 ± 0.02
Categorical Fidelity (*6)	96.7 ± 1.79
Privacy Risk (*7)	100.0 ± 0.00
Data Density (*8)	99.9 ± 0.09
Coverage Score (*9)	0.0 ± 0.00
Feature Accuracy (*10)	92.3 ± 2.24
Novelty Score (*11)	0.0 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	N/A	0.7025641025641025	0	
F1	0.7997903571549948	N/A	0.7997903571549948	0	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6820512820512821	N/A	0.6837606837606838	0.0017094017094017144	
F1	0.7891971402618019	N/A	0.7900741587708628	0.0008770185090608562	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6606837606837608	N/A	0.6487179487179489	-0.011965811965811956	
F1	0.7624201993972954	N/A	0.754312150221826	-0.008108049175469367	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	N/A	0.7068376068376068	0	
F1	0.8279091624529537	N/A	0.8279091624529537	0	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7034188034188034	N/A	0.7034188034188034	0	
F1	0.8108734553482932	N/A	0.8108734553482932	0	

Model: AB

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	N/A	0.6743589743589744	0	
F1	0.7821343746683586	N/A	0.7821343746683586	0	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	N/A	0.558974358974359	0	
F1	0.5587432119380917	N/A	0.5587432119380917	0	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	N/A	0.6863247863247863	0	
F1	0.7971299319768493	N/A	0.7971299319768493	0	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7076923076923076	N/A	0.6948717948717948	-0.012820512820512808	
F1	0.8136628637901065	N/A	0.8006162178062566	-0.013046645983849885	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7128205128205127	N/A	0.7042735042735043	-0.008547008547008527	
F1	0.81495475736121	N/A	0.8094311800077019	-0.005523577353508146	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	68	Female	0.6	0.1	1620	95	127	4.6	2.1	0.8
2	58	Female	1.7	0.8	1896	61	83	8.0	3.9	0.95
3	32	Male	12.1	6.0	515	48	92	6.6	2.4	0.5
4	60	Male	11.5	5.0	1050	99	187	6.2	2.8	0.8
5	56	Male	17.7	8.8	239	43	185	5.6	2.4	0.7
6	26	Female	0.6	0.2	142	12	32	5.7	2.4	0.75
7	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
8	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
9	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
10	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
2	45	Female	0.8	0.2	165	22	18	8.2	4.1	1.0
3	39	Male	3.8	1.5	298	102	630	7.1	3.3	0.8
4	68	Female	0.7	0.2	186	18	15	6.4	3.8	1.4
5	44	Male	0.9	0.2	182	29	82	7.1	3.7	1.0
6	50	Male	1.7	0.8	331	36	53	7.3	3.4	0.9
7	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
8	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
9	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
10	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	21	Male	18.5	9.5	380	390	500	8.2	4.1	1.0
2	62	Male	6.8	3.0	542	116	66	6.4	3.1	0.9
3	38	Male	1.5	0.4	298	60	103	6.0	3.0	1.0
4	55	Male	0.9	0.2	190	25	28	5.9	2.7	0.8
5	45	Male	2.2	0.8	209	25	20	8.0	4.0	1.0
6	35	Male	2.0	1.1	226	33	135	6.0	2.7	0.8
7	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
8	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
9	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
10	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	20	Female	0.6	0.2	202	12	13	6.1	3.0	0.9
2	43	Male	1.3	0.6	155	15	20	8.0	4.0	1.0
3	49	Female	0.8	0.2	198	23	20	7.0	4.3	1.5
4	38	Female	2.6	1.2	410	59	57	5.6	3.0	0.8
5	4	Male	0.8	0.2	460	152	231	6.5	3.2	0.9
6	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
7	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
8	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
9	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
10	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	23	Male	1.0	0.3	212	41	80	6.2	3.1	1.0
2	40	Male	0.6	0.1	98	35	31	6.0	3.2	1.1
3	12	Male	0.8	0.2	302	47	67	6.7	3.5	1.1
4	55	Female	8.2	3.9	1350	52	65	6.7	2.9	0.7
5	12	Male	1.0	0.2	719	157	108	7.2	3.7	1.0
6	27	Male	1.3	0.6	106	25	54	8.5	4.8	nan
7	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
8	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
9	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
10	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0
2	26	Female	0.6	0.2	142	12	32	5.7	2.4	0.75
3	50	Female	27.7	10.8	380	39	348	7.1	2.3	0.4
4	45	Female	0.8	0.2	165	22	18	8.2	4.1	1.0
5	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0
6	33	Male	1.5	7.0	505	205	140	7.5	3.9	1.0
7	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
8	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
9	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
10	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	60	Male	2.2	1.0	271	45	52	6.1	2.9	0.9
2	48	Male	2.4	1.1	554	141	73	7.5	3.6	0.9
3	74	Male	0.6	0.1	272	24	98	5.0	2.0	0.6
4	36	Male	0.8	0.2	158	29	39	6.0	2.2	0.5
5	35	Male	0.8	0.2	279	20	25	7.2	3.2	0.8
6	49	Male	1.1	0.5	159	30	31	7.0	4.3	1.5
7	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
8	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
9	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
10	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Male	16.4	8.9	245	56	87	5.4	2.0	0.5
2	52	Male	1.8	0.8	97	85	78	6.4	2.7	0.7
3	58	Male	0.9	0.2	1100	25	36	7.1	3.5	0.9
4	32	Female	0.7	0.1	240	12	15	7.0	3.0	0.7
5	50	Male	0.6	0.2	137	15	16	4.8	2.6	1.1
6	30	Female	0.8	0.2	158	25	22	7.9	4.5	1.3
7	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
8	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
9	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
10	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	51	Male	0.8	0.2	230	24	46	6.5	3.1	nan
2	45	Male	2.2	1.6	320	37	48	6.8	3.4	1.0
3	40	Male	3.9	1.7	350	950	1500	6.7	3.8	1.3
4	32	Male	15.6	9.5	134	54	125	5.6	4.0	2.5
5	40	Female	2.1	1.0	768	74	141	7.8	4.9	1.6
6	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
7	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
8	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
9	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
10	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

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B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	60	Male	1.9	0.8	614	42	38	4.5	1.8	0.6
2	35	Female	0.9	0.3	158	20	16	8.0	4.0	1.0
3	78	Male	1.0	0.3	152	28	70	6.3	3.1	0.9
4	18	Male	0.6	0.2	538	33	34	7.5	3.2	0.7
5	49	Male	1.1	0.5	159	30	31	7.0	4.3	1.5
6	51	Male	2.9	1.3	482	22	34	7.0	2.4	0.5
7	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
8	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
9	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
10	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3